Xu Yuanjian(徐元健)

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Education

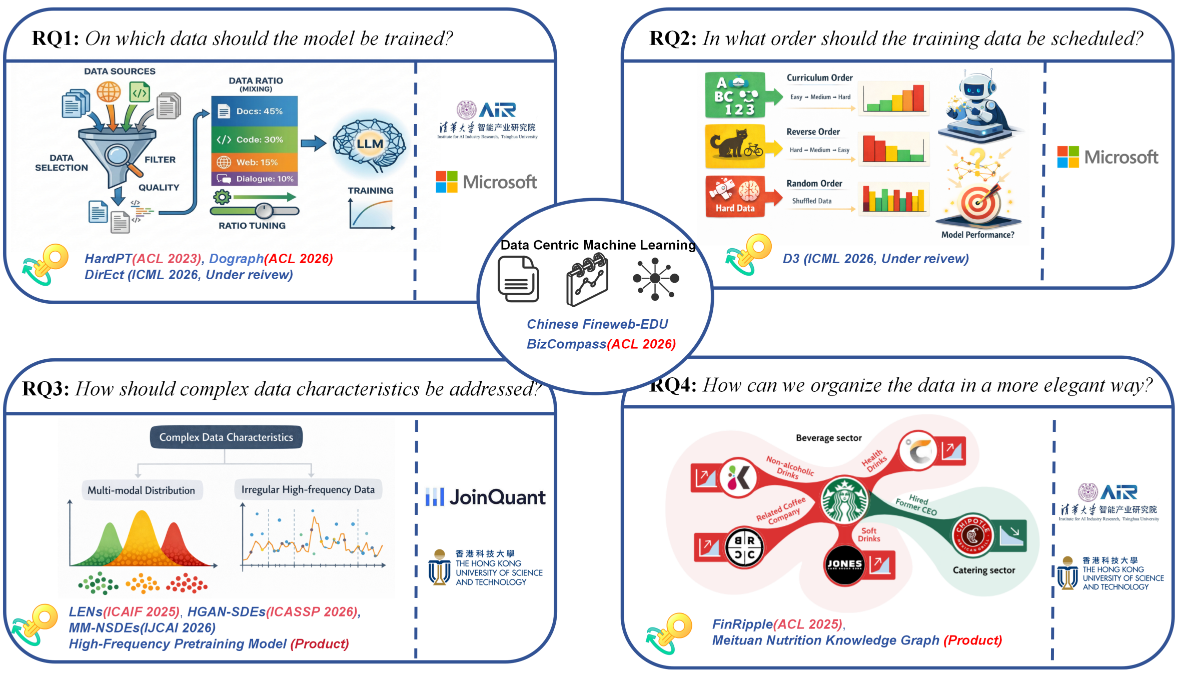
Hong Kong University of Science and Technology, Ph.D. in Fintech (Candidate) 2023.09–Present
Peking University, Master in Computer Science 2020.09–2023.06
Nankai University, Bachelor in Computer Science 2014.09–2018.06

A Brief Introduction

I am a Ph.D. candidate in Computer Science at the Hong Kong University of Science and Technology, Guangzhou campus. I am supervised by Prof. Guang Zhang from HKUST-GZ.
My research focuses on **Data Centric Artificial Intelligence**, particularly on understanding the mechanisms of **large language models (LLMs)**. My research systematically addresses four core data-centric questions:

RQ1: Which data to select for *LLM training*?,
RQ2: In what order to schedule training data to maximize performance?
RQ3: How to model *complex data characteristics* (e.g., irregular high-frequency and multi-modal data)?
RQ4: How to elegantly organize *heterogeneous data*?

To address these questions, I have published several papers at leading venues including **ACL 2023**, **ACL 2025**, **ACL 2026**, **ACM ICAIF 2025**, and **ICASSP 2026** (oral). Additionally, I have two manuscripts under review at **ICML 2026** (OA $>$ 4) and one at **IJCAI 2026** (passed first round). *Note: All publications and submissions are first-author works.* I also serve as a reviewer for leading conferences such as **NeurIPS**, **ICLR**, and **AAAI**.



Selected Publications (First Author)

- **Yuanjian Xu**, et al. "D^3: Dynamic Directional Graph-Constrained Data Scheduling for LLM Training." **ICML 2026 (Incoming)** *(CCF A, CORE A*; All reviews are positive)*.
- **Yuanjian Xu**, et al. "Towards Efficient LLMs Annealing with Principled Sample Selection." **ICML 2026 (Incoming)** *(CCF A, CORE A*; All reviews are positive)*.
- **Yuanjian Xu**, Jianing Hao, Guang Zhang, et al. "DoGraph: Towards Domain-Aware LLM Training with Graph-Guided Weighting." **ACL 2026 (Incoming)** *(CCF A, CORE A*)*.
- **Yuanjian Xu**, et al. "BizCompass: Benchmarking the Reasoning Capabilities of LLMs in Business Knowledge and Applications." **ACL 2026 (Incoming)** *(CCF A, CORE A*)*.

- **Yuanjian Xu**, Yuan Shuai, and Guang Zhang. "Hermite Discriminator for Neural SDEs." **ICASSP 2026 (Oral)** *(CCF B, CORE A)*.
- **Yuanjian Xu**, Jianing Hao, and Guang Zhang. "FinRipple: Aligning Large Language Models with Financial Market for Event Ripple Effect Awareness." **ACL 2025** *(CCF A, CORE A*)*.
- **Yuanjian Xu**, Jianing Hao, and Guang Zhang. "LENS: Large Pre-trained Transformer for Exploring Financial Time Series Regularities." **ICAIF 2025** *(Leading conference for AI in finance)*.
- **Yuanjian Xu** and Zaiqing Nie. "Hard Sample Aware Prompt-Tuning." **ACL 2023** *(CCF A, CORE A*)*.

Under Review Works

- **Yuanjian Xu**, Jianing Hao, Guang Zhang. "State-Aware Neural Stochastic Differential Equations for Multi-Modal Dynamics." **Under review at IJCAI 2026** *(CCF A, CORE A*; passed first round)*.
- H. Ding*, Jianing Hao*, **Yuanjian Xu**, et al. "A^4: Tree-Based Action Advantage Attribution for LLM Agent Evolution." **Under review at ACL 2026** *(CCF A, CORE A*; *Equal contribution)*.
- Jianing Hao, H. Ding, **Yuanjian Xu**, T. Sun, R. Chen, W. Zhang, G. Zhang, S. Li. "Game-Theoretic Lens on LLM-based Multi-Agent Systems." **Under review at IJCAI 2026 (Survey track)** *(CCF A, CORE A*)*.

Research & Teaching Experience

Hong Kong University of Science and Technology, Teaching Assistant, Advanced Statistics

INTERNSHIP EXPERIENCE

AIR, Tsinghua University Research Intern | 2022.03–2023.03

Worked under the supervision of Prof. Zaiqing Nie. Contributed to the *Meituan Nutrition Knowledge Graph* construction. Investigated *hard sample problems* in NLP and proposed *HardPT*, published at *ACL 2023*.

Microsoft Research Asia (MSRA) Research Intern | 2024.10–2025.02

Worked under the supervision of Dr. Zhong Li. Focused on *data selection* and *training order optimization* for *large language models*. Proposed the *D^3 method* and an *annealing training framework*, both currently under submission.

Joinquant (Billion-scale quantitative fund) Research Intern | 2025.02–2025.05

Worked under the supervision of PM Ruixiao. Developed *tick-level generative and representation models* for *high-frequency trading*. Addressed key challenges including *non-equally spaced data* and market randomness.

Skills

Machine Learning: I am well-versed in modern generative models, LLMs, and machine learning.
Mathematical Foundation: I have a relative solid mathematical foundation. During my Ph.D. studies, I passed the qualifying exam in *optimization*, *advanced probability theory*, *advanced statistics*, and *stochastic analysis*.
Financial Theory: I also have exposure to financial theory, having passed *FRM Level 1* and studied *advanced microeconomics* and *advanced asset pricing*.

Honors & Awards

- Award for Excellent Academic Excellence, Peking University 2021 Certificate No.: H2021000170320
- Air Star Plan, Tsinghua University, Institute for AI Industry Research (AIR) 2021
- Full Ph.D. Scholarship, Hong Kong University of Science and Technology 2023–Present
- Tomorrow Star Plan, Microsoft Research Intern 2025